

How We Classify Downwind and Upwind AC Area and Population

Corrected East-referenced construction

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Executive summary

For each focal grid and month, one rolling wind direction divides its assigned assembly constituency (AC) into a downwind half and an upwind half. We then construct two related but distinct indicators:

$$\text{downup_ac_area} = \mathbf{1}[A_{\text{down}} > A_{\text{up}}], \quad \text{downup_ac_pop} = \mathbf{1}[P_{\text{down}} > P_{\text{up}}].$$

The area indicator compares exact polygon areas. The population indicator compares the sum of 2010 population assigned to other small-grid centroids in the same AC. Therefore, the indicators can differ when population is unevenly distributed across the AC.

One angular convention throughout

Every directional calculation is a Cartesian angle measured counterclockwise from East:

$$0^\circ = \text{East}, \quad 90^\circ = \text{North}, \quad 180^\circ = \text{West}, \quad 270^\circ = \text{South}.$$

The wind is $w = \text{atan2}(v_{10}, u_{10})$. For a comparison-grid centroid j relative to focal grid i , the bearing is

$$\beta_{ij} = \text{atan2}(y_j - y_i, x_j - x_i).$$

No step interprets either angle as a clockwise-from-North compass bearing. The normalized calculation angle remains East-referenced; normalization only changes its numeric range to $[0, 360)$.

Real example used in the figure

The reproducible script selected grid **77902** in **Prithla** AC (AC ID **262**), Faridabad, Haryana, for **2019-03**. The rolling wind direction is **-57.820° from East**.

Measure	Downwind	Upwind	Indicator
Area (km ²)	186.74	99.54	1
Area share (%)	65.2	34.8	
Population	115438	152215	0
Population share (%)	43.1	56.9	

Here, the downwind side contains the majority of AC area but not the majority of the comparison-grid population. Thus `downup_ac_area=1` and `downup_ac_pop=0`.

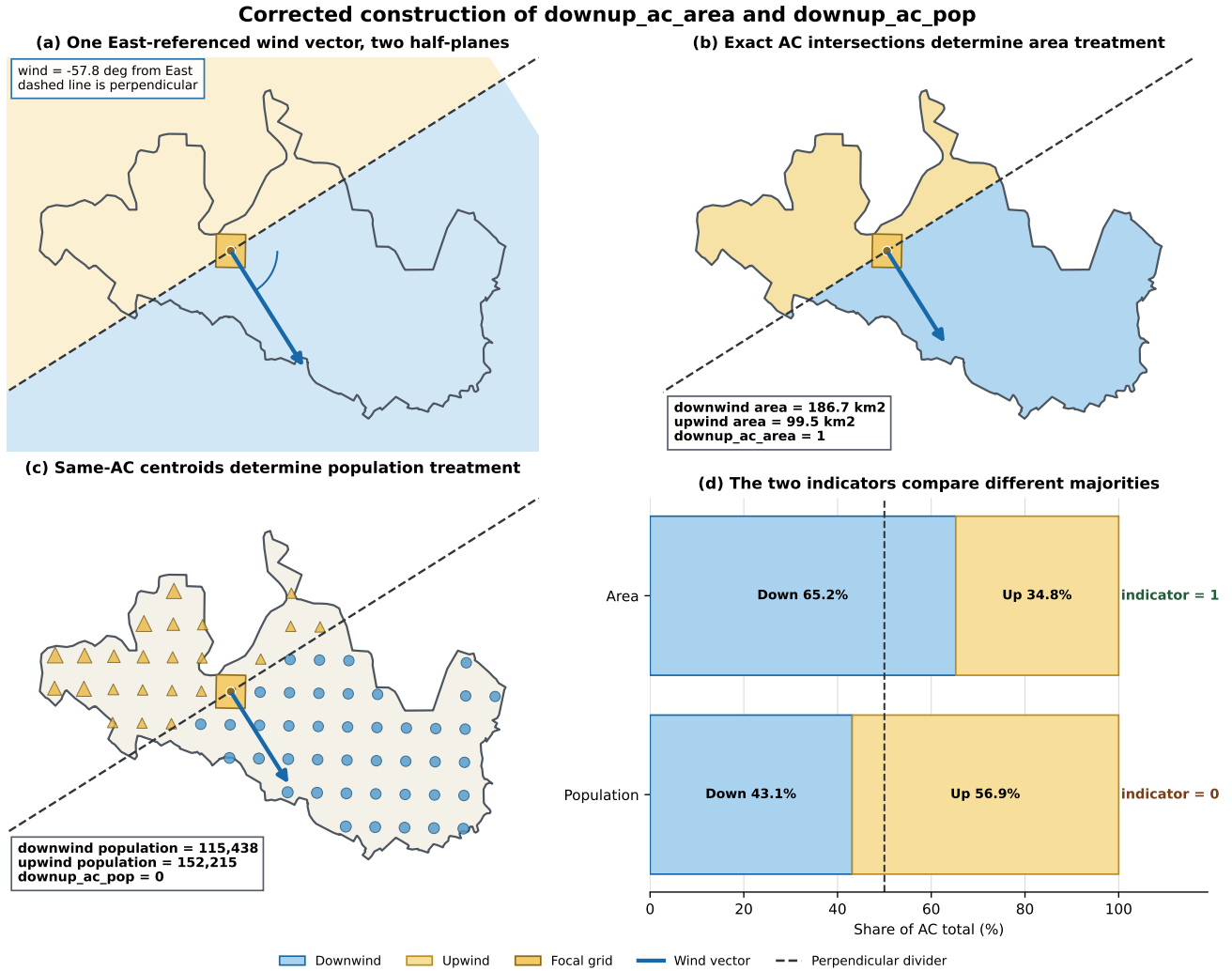


Figure 1: Corrected construction for one real grid-month. Panel (a) uses the East-referenced wind vector and a perpendicular line through the focal-grid centroid. Panel (b) intersects the AC with the two complementary squares. Panel (c) classifies every other same-AC population centroid using the same wind direction; marker size represents population. Panel (d) shows why the two majority indicators can differ.

Step 1: construct the area measure

Let $p = (x_i, y_i)$ be the focal-grid centroid and let θ be the rolling wind direction from East. Define

$$v = (\cos \theta, \sin \theta), \quad t = (\sin \theta, -\cos \theta),$$

where v points downwind and t is perpendicular to the wind. Choose R larger than the maximum distance from p to the AC bounds. The downwind square has corners

$$p - Rt, \quad p + Rt, \quad p + Rt + 2Rv, \quad p - Rt + 2Rv.$$

The upwind square uses $-v$. We compute the exact intersections

$$A_{down} = \text{area}(AC \cap S_{down}), \quad A_{up} = \text{area}(AC \cap S_{up}).$$

The binary indicator equals 1 only when $A_{down} > A_{up}$. An exact tie is coded 0. A missing rolling direction produces a missing indicator.

Step 2: construct the population measure

For every other grid j assigned to the same AC, compute the signed circular difference

$$\delta_{ij} = ((\beta_{ij} - \theta + 180) \bmod 360) - 180.$$

Grid j is downwind when $|\delta_{ij}| < 90^\circ$ and upwind when $|\delta_{ij}| > 90^\circ$. The focal grid is excluded because its centroid is on the dividing line. A comparison centroid exactly on that line is also excluded from both sums. Population is then

$$P_{down} = \sum_{j: |\delta_{ij}| < 90^\circ} Pop_{j,2010}, \quad P_{up} = \sum_{j: |\delta_{ij}| > 90^\circ} Pop_{j,2010}.$$

The binary indicator equals 1 only when $P_{down} > P_{up}$; ties are 0 and missing rolling direction produces a missing value.

What is shared and what is different

Component	Area construction	Population construction
Focal unit	grid centroid	grid centroid
Wind angle	rolling, from East	rolling, from East
Divider	perpendicular line	same perpendicular line
Objects classified	AC polygon surface	same-AC grid centroids
Weights	square kilometres	2010 grid population
Indicator	down area > up area	down pop. > up pop.

Implementation safeguards

The production code retains the unrounded rolling-direction float for every grid-month. Area rows are never rounded, grouped, cached, or deduplicated by wind direction. The population lookup uses circular intervals and an ASOF join only after both the wind and centroid bearings have been placed in the same East-referenced $[0, 360)$ system.

The automated tests enforce the four cardinal cases:

- East wind: the eastern half-plane is downwind.
- North wind: the northern half-plane is downwind.
- West wind: the western half-plane is downwind.
- South wind: the southern half-plane is downwind.

They also test the $179^\circ / -179^\circ$ wraparound to ensure that monthly and rolling directions are calculated from eastward/northward vector components, not arithmetic averages of degree values.

Production sequence

The corrected files are generated in this order:

1. `build_wind_direction_grid_duckdb.py`
2. `build_downup_ac_pop_cluster.py`
3. `build_downup_ac_area_cluster.py`
4. `build_downup_13kmpl_cluster.py`
5. `build_0_master_dataset.py`
6. `build_all_stacked_datasets_duckdb.py`

The one-job launcher is `rebuild_corrected_downup_pipeline.sbatch`.

Interpretation

Both indicators describe exposure generated by a hypothetical fire at the focal grid under the prevailing rolling wind. The area measure asks whether most of the constituency's land lies downwind. The population measure asks whether most of the measured same-AC population lies downwind. Neither measure should be interpreted as the other, and disagreement between them is a valid feature of the construction rather than a merge error.